

A Packing of 62 Lines in $\mathbb{R}^{13}$

Limin Ge*

October 8, 2026

Abstract

We give 62 1-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 0.907785. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.907084, obtained by deleting 1 of the 63 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 62 subspaces of dimension 1 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.907785 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 1$. The cell $N = 62$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.907084, obtained by deleting 1 of the 63 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(13 - 1)/13 \cdot 62/61 = 0.938209$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n1N62.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 70 seconds on one laptop, with about ten such searches running at once.

A Exact value

```
numerator:
28368301228900150881574919426307464371030067633
denominator:
31249999999941852061477994120726020629411829433
```

B Coordinates

62 blocks of 13 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m13n1N62.txt`.

321440277782	86592543975	-350813610983	-112284995196
25349725546	-246232359198	-185980241857	225146375176
-532775802662	-423669134684	-225905441578	-223234146189
-485213594462	197525590573	563094123443	-17282593091
-165282041978	-241355633737	113536808949	-88955116507
357273169457	-444734353898	453740375759	392059714316
310390094648	-5791215132	-65842246516	693324242368
51826430234	403588850878	177885818431	330111675027
171710382271	176371200913	-275693999150	220201885515
50140735172	283280824309	-138555690293	-56409485464
91474151356	74533366209	125534128723	-189195753251
-211231636967	415392206312	-119453856782	328762566283
194360790323	187669885668	47524652665	319997793777
307215908636	-397395526817	127181027280	93284680942
-444013633398	-35380172881	144985926091	383236078852
227259712471	-328800187512	513310899756	-119263885659
-506375300567	573740896335	208827864658	-168517573564
3461315673	130653643689	-25787545950	-373784411553
-4000787164	88126927481	-631068898597	118043630304
-203330300466	226996799023	165238699549	50748054847
-502555164272	49918561791	-128962545331	282248032240
21139094327	137002319335	-190841504687	-79639036492
-78403222045	-480963176932	310444639810	-406039384708
216363444895	-297551705606	121118249618	-429213066341
224124441212	55993851231	-96852041957	-662496388276
-53250668540	326521421726	-320485446291	374023912425
-193300377893	198034820607	-405982348244	183702576278
-247574838199	131398347657	214936025022	-38631316344
35640970650	-208606695298	207283426185	76481859157
694906294366	433451264402	85129575456	26592660901
84844771021	-428857455321	-28077750121	314451273145
393373959074	236602288598	102546316501	-203112215372
55227170624	-141944111511	7645155759	427665064109
-128737088814	-60734344317	51037058958	86510092237
347961351259	-196411655134	172050056435	-173259569299
51046725342	523780949879	396197592331	39423244976
-314422091324	-336402470415	99656177077	131606762730
38299452068	153016526632	-57117268777	-157995059920
107706414479	238499906303	44188144876	-205986863542
70961154676	-540636275751	-411716492810	384013318221
12011233170	49165808353	-267952294791	-176114232267
224058017889	-196521650803	731953983195	339677154181
326918900731	-303325106081	-410448450825	-214222291900
-356601457706	-166072505062	349261386738	-296094340107
-301575983512	49540703113	256993012278	183661267778
565215908034	119455678670	341316785803	607190946949
-48916659929	-154846173637	-531673759471	-132010359784
241333365054	247093643483	-27182548979	-92097699957
158703206684	13861496332	-13090496923	-39395944347
-407341877852	-603877095779	301486003369	274422730201
-126840092528	35980725494	-12399314998	16003291453
-179787181304	-152618054814	-268796911859	401311497171
-433192576816	-192226583115	-30079197332	42563363883
178673437486	-313066522255	-276285212270	83869148506
-771692100	-526955762088	53676326578	-61660698989
145989229232	54334413629	101964105041	101859242613
583644686011	-22701820641	28248447352	-174246174072
394347250771	341704811407	159149241709	266115511952
268058449958	-46711720503	-277329826240	-435382506807
-26771293672	120176180062	29821715881	-388663965117
-288990824988	549843723619	665602393431	271011474636
-260678342977	-43547903244	132628366360	-344659367487
153558713099	287050410316	6147608332	47788454654
-38517008440	-368335517088	-95198118015	51840747151
-117689852479	-19935257549	64260027800	-396484765767
-156580777399	59645403311	124243916112	286597113930
66573492044	224116405950	430249881516	435599423694
-339600605133	22412160507	-459649356829	119238545301
-77072469432	-261496348481	-191276097886	264200054866
379713667852	382032147839	-105926781590	147225249940

-34051969637	-36311349253	-256277005533	473674111471
-495106012905	183220052662	382203802445	356423030594
205732030480	237707618779	-184391864249	309471066592
200216557556	60988982931	95535142783	202402184761
-343875942951	33208363698	-399756095294	427725222303
124821179456	668233097027	148784038240	238300156820
530803240196	-262160085722	-337278796970	16119267710
-330866393838	380494985571	441567172203	94285836041
125974628651	49484323624	-223620991265	414566045198
261424912599	343307094531	301377561776	37867780715
308248716759	-76083564551	-234684581644	-28782749701
-69657973805	-182098519270	-123087218927	93291222685
212084814527	-399128528097	-54070362444	248351844843
405612574794	192488370675	-243125961165	-175163658191
329960470124	13538729728	261207615207	69505016552
86945811638	-444706797668	521972718818	184102575388
-101132796256	314522401989	-15717374417	-36868710985
251088105587	500647858212	-261649768648	119564659191
-159640716301	64708749917	-159239811132	-276737582163
71544694380	-34800348437	595388043804	-36943837713
-113551928314	-302438865414	-217600176343	51365810684
-357992845645	-156413568299	-108315481740	595603211109
218034744094	18154920425	139355485646	394729391298
277281591110	145939081705	-105662598878	510043710270
-325379590132	86572480752	-121981523619	44421751285
-640168068414	-282615501205	51541290481	-379002657459
252355273751	-59241134942	119775714886	363204326091
116831052313	-379088837283	-283770549228	34083313482
70709296045	-470129423858	-256636893419	75281272141
-188763100373	473097257614	-74004417724	-346267365984
309431311793	-378706871680	-47799660528	150272225179
2333785810	-105185332335	-584882628572	186502093537
-99684459030	-273005954688	-111153727732	117654617223
-548810987851	-213225265740	488434599579	-208440246261
-502251201601	113788077851	333019938595	-472674383036
115369134360	73448570008	314467864127	-178418368682
-115597770694	-287004231759	-15218212189	455645486253
124587920138	-287603521427	-413321060800	-142083984443
-147294150599	-366634737287	220685681737	304936716473
183557790025	291308403943	123889790580	39500461905
-107443104330	-269940226445	-22533510227	529580648563
210083667104	-198591549025	331722042	267722991842
90187934194	-226607214486	289778635978	-5180391877218
-39386471089	107904662394	-162032820561	272515747592
523129862104	-253535291095	-315894622210	-188459422046
63766689566	518527022552	-478591362245	-320686791657
325924477610	298056634813	12387746055	-204169381099
-321941037247	24896310325	35888526425	158620828
135652360222	352668255508	-569094497877	-43828874971
93473656806	333214018113	-443701168613	126793996274
67963459296	-677528411792	-20440456993	-112484645041
625933662592	39417757973	47546827397	-151704236522
102695930494	380893492251	-364899071449	-200320521025
328673125687	-156219815426	227244122664	-241569287855
-128617696407	23706404129	-433991297996	-246420831150
253033894546	74831126653	-265557430725	190538018998
240337457421	-135266224713	-225188878157	73990863168
-324972166300	-217129991950	-54671575425	291267837310
416221168304	-98188055913	370091046352	-223551225950
61892106155	229446576426	110068706810	-112534600070
144319535518	299782099959	357519491211	300765469310
-68198003123	1203360925	161829042836	716174132408
-56739582991	56651714407	-390916039322	-20265864765
264684828236	-79807386840	-344712311891	-157312051283
341014433464	-156701029550	152615766554	-138664297426
-345152696592	-416962318474	-464291867885	-156510955912
382225937375	233206485000	-149623267467	233542695440
-279470734104	-101592254892	-208580596662	148435053899
-333251105374	-215371787831	162109882130	-287621434048
367191414401	86310634297	39693546510	32337526094
136620464985	-113403041235	183806543402	312756763
109154086508	698153690633	304519940793	-512149230885
38142872198	-288648942569	-173970023510	234381505897
286601143767	-274294303187	-475286422990	271102366107
-515618462588	12048191203	-114330178691	254333136923
-5779549307	-39452414975	-41429086012	-129733401229
-73144803196	206017287758	170293915323	568933525908
395183528633	33543342102	-357840720852	-13224889366
150377905041	146287698318	270313613117	-289619834811
419230013018	52171718657	239127514291	448860519544
355582500798	-773687663304	9403573402	1672265591
218096080351	-94049034222	447507750138	541552392808
307904206486	-20398934143	-462447987565	-173046868952
99484530214	70194990760	55311077347	404844310938
-242671148758	474826609599	198482752458	-385027586041
146540144682	-130878348505	-30655415660	33862787545
4541148679	-204557367639	-498797788917	173821935627
76549138274	-82240805363	67666478718	-170557083914
307055794617	136150577104	-44617560910	86641394064
-167453537662	284676534729	366343356358	105547696624
243630520231	-106487688123	185258200229	-374929920787
62737506564	-270272537333	-50453231078	-16066307155
-48271698574	128000221090	641395200050	83009601298
622795939451	320469249505	-175801969533	-449612412249
-94933414677	166962062478	-168319507099	-64262916368
87753423370	155888498155	102152871752	141119733935
-566765011216	-155867961817	-214771493193	85339950156
251153528331	571039804091	183509902321	306872391796
160149581341	495766802065	-303586377617	299153712006
-176715172459	-32888911324	-398787412257	-419376639194
349266841915	296114268017	121967704865	441591116196
-42463583759	157590367169	283090657156	-97677338230

-233914843985	-152105963058	-548867740484	-243984477688
7344096933	-7948121108	-362400287946	265810738363
149805743403	227974704949	295227321332	473334593564
-587761338795	-280282164548	153959909507	-139507927046
-296772666742	85606992629	-270492560360	-97649741887
-79575127699	514162007009	-241910143500	-42798023230
-201700009920	-134663988819	-324468476540	280470255030
-212670358170	383639710952	486745593868	-501900654722
-31715374549	1103783308	13076901956	-73636773286
-193954860455	-133678645162	191813540174	-305030151501
333279940081	-138638138175	90625052858	75464047725
-448017613795	449531605988	422202186321	-275364215586
-269885042923	178779161728	-559473419815	-375042237273
-168717308479	64513082011	-64415430318	-140139993577
170668910080	408876154823	67303088217	-272534379132
-6868059219	-145713164060	130930299197	287024746638
-532208813718	-143869785247	-184285124181	-173214725570
312645270522	504912029848	174927904203	399815019585
357750106190	134877580685	-175996664591	120840155656
453420497873	71914896463	403603458459	501376156529
-115816428698	413125406508	654398816428	409072988751
207534399077	286646096305	-54537824918	-376032263597
12468637666	-48247965275	-27844529101	-92443283170
294841347391	26252291763	-370071555982	-24385480713
173014106400	-63750116240	-192319216817	-286991983002
-116163879188	215714507996	167758047679	-200129114432
-263168374033	14266268721	-210115687180	-30004407010
576701628161	-164628734107	-40502511347	-143468809737
221615183255	-494478507931	-132550226571	
98467483401	41946592148	-97426654440	